

Name: _____ Date: _____ Period: _____

1. A student needs to find the **vertical asymptotes and holes** of a rational function. Circle the form that is most useful, and explain why in one phrase.

A. Factored form **B.** Standard form

Because: _____

2. Perform the first step of polynomial long division to divide $x^2 + 3x + 5$ by $x + 2$.

What is the first term of the quotient? _____

After multiplying and subtracting, what is the new dividend? _____

3. Use Pascal's Triangle to expand $(x + 2)^4$. (Row 4 of Pascal's Triangle: 1 4 6 4 1)

$(x + 2)^4 =$ _____